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2024-CET-04

Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL DDL Commands (Database & Table Creation)

Submitted To:

Engr. Asma Javid

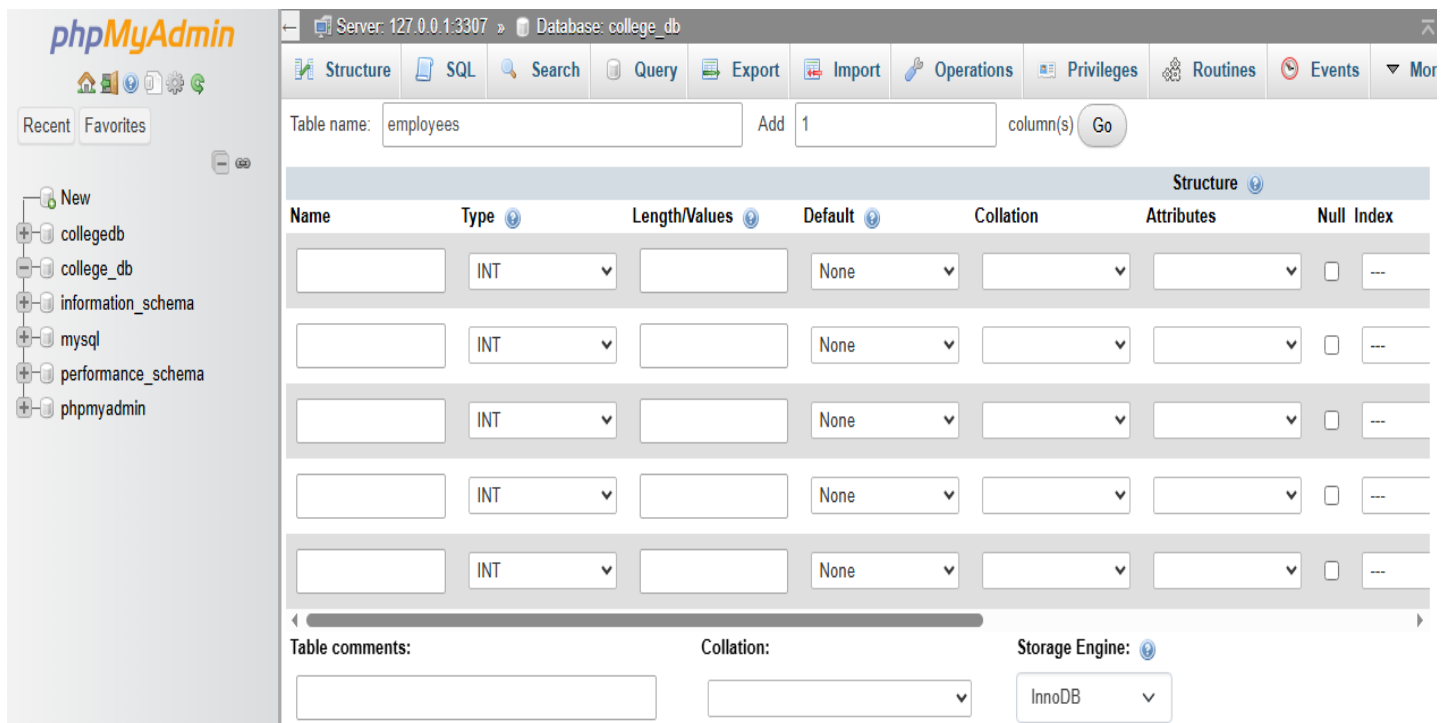
Lab Task 4

Tasks (Practice Work)

1.Create Database:

Create a database named:

college_db



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases: 'collegedb', 'college_db' (selected), 'information_schema', 'mysql', 'performance_schema', and 'phpmyadmin'. The main area is titled 'Server: 127.0.0.1:3307 » Database: college_db'. Below the title bar are tabs for 'Structure', 'SQL', 'Search', 'Query', 'Export', 'Import', 'Operations', 'Privileges', 'Routines', 'Events', and 'More'. The 'Structure' tab is active, showing the 'Table name: employees' and 'Add 1 column(s) Go' buttons. Below this is a table structure editor with columns: Name, Type, Length/Values, Default, Collation, Attributes, Null, and Index. There are five rows, each with an empty 'Name' field, 'INT' type, empty 'Length/Values' field, 'None' default, empty 'Collation' field, empty 'Attributes' field, and 'Null' checkbox. At the bottom, there are fields for 'Table comments:', 'Collation:', and 'Storage Engine: InnoDB'.

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---

Table comments:

Collation:

Storage Engine:

2.Create Table:

Create a table named employees with following fields:

Column Name	Data Type	Constraint
emp_id	INT	PRIMARY KEY, AUTO_INCREMENT
emp_name	VARCHAR(50)	NOT NULL
salary	DECIMAL(10,2)	DEFAULT 25000
department	VARCHAR(30)	
join_date	DATE	DEFAULT CURRENT_DATE

The screenshot shows the phpMyAdmin web interface. The top navigation bar includes links for Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, and a More menu. The left sidebar shows a tree view of databases, with 'college_db' selected. The main area displays a text editor for SQL queries. The query entered is:

```
1 CREATE TABLE employees
2 (
3   emp_id INT AUTO_INCREMENT PRIMARY KEY,
4   emp_name VARCHAR(50) NOT NULL,
5   salary DECIMAL(10,2) DEFAULT 25000,
6   department VARCHAR(30),
7   join_date DATE DEFAULT CURRENT_DATE
8 );
```

Below the query editor are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters'. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is at the bottom right.

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

SELECT * FROM `employees`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

emp_idemp_namesalarydepartmentjoin_date

Query results operations

Create view

Console

3.Insert Data:

Insert at least 5 records into employees.

The screenshot shows the phpMyAdmin interface with the 'employees' table selected in the 'college_db' database. The SQL query editor contains the following code:

```
1 INSERT INTO employees (emp_name, salary, department)
2 VALUES
3 ("Zara Khan",50000,"Software Engineering"),
4 ("Zoya Nayak",50000,"Software Engineering"),
5 ("Bilal Khan",55000,"Software Engineering"),
6 ("Saifullah Khan",530000,"Electrical Engineering"),
7 ("Toseef Khan",100000,"Medical Engineering");
8
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is at the bottom right.

The left sidebar shows the database structure with 'college_db' expanded, showing the 'employees' table. The top navigation bar includes links for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', 'Privileges', 'Operations', and 'Triggers'.

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

SELECT * FROM `employees`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

emp_id

emp_name

salary

department

join_date

1

Zara Khan

50000.00

Software Engineering

2026-04-02

2

Zoya Nayak

50000.00

Software Engineering

2026-04-02

3

Bilal Khan

55000.00

Software Engineering

2026-04-02

4

Saifullah Khan

530000.00

Electrical Engineering

2026-04-02

5

Toseef Khan

100000.00

Medical Engineering

2026-04-02

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Console

4. Modify Table:

1. Add column email

2. Add column phone

3. Set email as **UNIQUE**

The screenshot shows the phpMyAdmin interface with the 'employees' table selected in the 'college_db' database. The SQL query editor is open, displaying the following queries:

```
1 ALTER TABLE employees ADD email VARCHAR(15) UNIQUE;
2 ALTER TABLE employees ADD phone VARCHAR(15);
3
```

The table structure on the right shows the following columns: emp_id, emp_name, salary, department, and join_date. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters' and a 'Go' button at the bottom right.

The screenshot shows the phpMyAdmin interface with the 'employees' table selected. The results of a query are displayed in a green box, indicating that 5 rows were shown. The query executed was:

```
SELECT * FROM `employees`
```

Below the results, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'.

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

courses

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 5 (6 total, Query took 0.0010 seconds.)

SELECT * FROM `employees`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

emp_id

emp_name

salary

department

join_date

email

phone

☐	Edit Copy Delete	1	Zara Khan	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐	Edit Copy Delete	2	Zoya Nayak	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐	Edit Copy Delete	3	Bilal Khan	55000.00	Software Engineering	2026-04-02	NULL	NULL
☐	Edit Copy Delete	4	Saifullah Khan	530000.00	Electrical Engineering	2026-04-02	NULL	NULL
☐	Edit Copy Delete	5	Toseef Khan	100000.00	Medical Engineering	2026-04-02	NULL	NULL
☐	Edit Copy Delete	6	Muhamamd Bilal	25000.00	Computer Engineering	2026-04-02	muhamamdbilal@g	305982154

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Console

Use:

DESCRIBE employees;

The screenshot shows the phpMyAdmin web interface. On the left is a sidebar with a database tree containing 'college_db' and its table 'employees'. The main panel has a top navigation bar with tabs: Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Triggers. The 'Structure' tab is active, displaying the table structure for 'employees'. A message at the top states 'Your SQL query has been executed successfully.' Below this, the executed query 'DESCRIBE employees;' is shown. Under the 'Extra options' section, a table lists the table's fields and their properties.

Field	Type	Null	Key	Default	Extra
emp_id	int(11)	NO	PRI	NULL	auto_increment
emp_name	varchar(50)	NO		NULL	
salary	decimal(10,2)	YES		25000.00	
department	varchar(30)	YES		NULL	
join_date	date	YES		curdate()	
email	varchar(15)	YES	UNI	NULL	
phone	varchar(15)	YES		NULL	

Below the table, the 'Query results operations' section contains buttons for 'Print', 'Copy to clipboard', and 'Create view'. At the bottom left of the main panel is a 'Console' tab.

6. Extra Challenge (optional):

Create table courses:

course_id, course_name, duration, fee

With proper data types and constraints.

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'college_db' selected and 'employees' as the current table. The top navigation bar includes 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', 'Privileges', 'Operations', and 'Triggers'. The main area is titled 'Run SQL query/queries on table college_db.employees:' and contains a text editor with the following SQL query:

```
1 CREATE TABLE courses
2 (
3   course_id INT AUTO_INCREMENT PRIMARY KEY,
4   course_name VARCHAR(30) NOT NULL,
5   duration VARCHAR(20) NOT NULL,
6   fee INT DEFAULT 25000
7 );
```

Below the query editor are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters'. At the bottom, there is a 'Delimiter' dropdown set to ';', checkboxes for 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked), and a 'Go' button. A 'Console' tab is visible at the bottom left.

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds.)

```
SELECT * FROM `courses`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

course_id	course_name	duration	fee
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Query results operations

Create view

Console

phpMyAdmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.courses:

```
1 INSERT INTO courses(course_name,duration,fee) VALUES
2 ("WEB DeveLopment","6 Months",50000);
```

course_id	course_name	duration	fee
-----------	-------------	----------	-----

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

courses

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 0 (1 total, Query took 0.0018 seconds.)

SELECT * FROM `courses`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

course_id

course_name

duration

fee

1

WEB Development

6 Months

50000

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Console